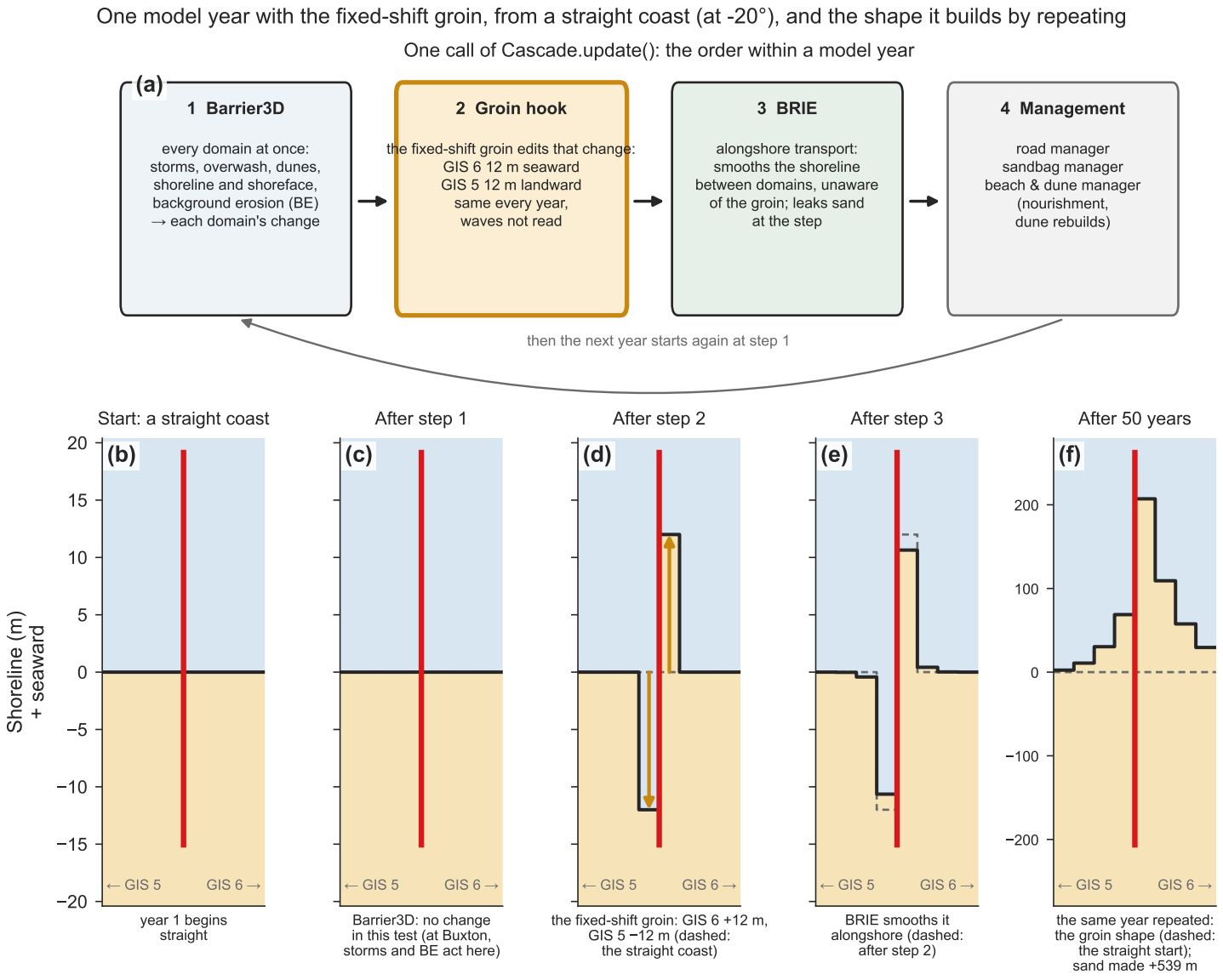
One model year with the fixed-shift groin, from a straight coast (at -20°), and the shape it builds by repeating
One call of Cascade.update(): the order within a model year
(a)
1 Barrier3D
every domain at once:
storms, overwash, dunes,
shoreline and shoreface,
background erosion (BE)
→ each domain's change
2 Groin hook
the fixed-shift groin edits that change:
GIS 6 12 m seaward
GIS 5 12 m landward
same every year,
waves not read
3 BRIE
alongshore transport:
smooths the shoreline
between domains, unaware
of the groin; leaks sand
at the step
4 Management
road manager
sandbag manager
beach & dune manager
(nourishment,
dune rebuilds)
then the next year starts again at step 1
Start: a straight coast
After step 1
After step 2
After step 3
After 50 years
Shoreline (m)
+ seaward
(b)
(c)
(d)
(e)
(f)
← GIS 5
GIS 6 →
year 1 begins
straight
Barrier3D: no change
in this test (at Buxton,
storms and BE act here)
the fixed-shift groin: GIS 6 +12 m,
GIS 5 −12 m (dashed:
the straight coast)
BRIE smooths it
alongshore (dashed:
after step 2)
the same year repeated:
the groin shape (dashed:
the straight start);
sand made +539 m